Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/01/2006
PI Name:	Meystre, Pierre Ph.D.		
Project Title:	Atom optics in controlled and microgravity environments		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	pierre.meystre@optics.arizona.edu	**Fax:**	FY 520-621-4358
PI Organization Type:	UNIVERSITY	**Phone:**	520-621-4651
Organization Name:	University of Arizona		
PI Address 1:	Optical Sciences Center		
PI Address 2:			
PI Web Page:	http://www.optics.arizona.edu/		
City:	Tucson	**State:**	AZ
Zip Code:	85721	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
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Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
Contact Email:	lute.maleki@jpl.nasa.gov		
Flight Program:			
Flight Assignment:			
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COI Name (Institution):			
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Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth's gravity		

Task Description:	The goal of this project is to carry out theoretical investigations of atom optics in environments characterized by controlled and adjustable accelerations and in field-free situations. The main thrusts of the research are: the theoretical study of nonlinear atom optics, and in particular matter-wave mixing phenomena, in accelerated frames; the investigation of the evolution of matter-wave coherence in accelerated atomic de Broglie waves, with applications in rotational and gravitational sensors; the detailed theoretical analysis of atom holography in microgravity environments, including the use of accelerated beams for the reconstruction process; and a broad theoretical investigation of field-sensitive situations that cannot realistically be performed in earhtbound laboratories. Atom optics is on the verge of becoming an engineering science with broad potential applications, in particular in rotation sensors, gravimeters and gravity gradiometers. Other possible applications include nanofabrication, atom holography, etc. In many cases, the atomic source will be a quantum degenerate gas, such as a Bose condensate --- the atom optical analog of a laser --- or, in the future, a Fermi-degenerate atomic system. Practical devices require however matter waves of short wavelengths, which can be realized by particle acceleration. In addition, external fields used e.g. to compensate earth gravity can lead to random frequency shifts and decoherence that must be avoided in precision measurements. For these and other reasons, it is important to study atom optics in controlled and microgravity environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Our work has continued in two major directions: the dynamics of BECs in double-well potential,s and the application of ultracold molecules to matter-wave interferometry, gravimeters, and accelerometers. Schemes involving ultracold atoms for the detection of weak forces with an accuracy beyond the standard quantum limit are now considered to be prime candidates for increased sensitivity. In particular, the so-called “contrast resonances” that appear when these forces modify the potential have been proposed by Kasevich and coworkers as a sensitive tool to measure these forces. Following these observations, we have now extended our previous results to the study of Bose-Einstein condensates in asymmetric double-well potentials subjected to sudden changes, and assessed the potential use of characteristic signatures of their dynamics to the detection of weak forces. We have concentrated on observable quantities such as fringe visibility and phase noise, that is, on the dynamics of the coherences of the matter-wave field, both first and second order, as functions of holding time after a sudden change of the Hamiltonian parameters. Specifically, we have used a combination of analytical and numerical techniques to investigate the possible sub-shot noise estimation of applied force fields through detection of the coherence dynamics of Bose-condensed atoms in asymmetric double-well traps, taking into account the full quantum dynamics todemonstrate the dynamical production of phase squeezing beyond the standard quantum limit. As already mentioned in the 2004 Taskbook, a substantial fraction of our efforts is now directed toward ultracold molecules, and in particular ultracold polar molecules. In this ocntext, we have now carried out a number of studies related to the formation of molecular dimers from ultracold atoms, either via photassociation or via the use of Feshbach resonance. We have characterized the quantum fluctuations in the time of formation of the molecules, so so-called "passage time statistics," and also determined the role of quantum fluctuations in processes that bring closed-channel Feshbach molecules from the weakly bound excited state in which they are produced to their ground electro-vibrational state, using for instance cavity-QED techniques. (A paper on this results in=s in preparation.) On a more practical side perhaps, we have also investigated possible molecular beam splitters that will allow the injection of ultracold polar molecules in two coherent counterpropagating beams inside a molecular storage ring. Our efforts focus now on the study of the study of the properties of ultracold molecular trapped in the storage rings which will eventually be used as rotation and/or acceleration sensors. In a recent first step toward this goal we have studied the Thomas-Fermi ground state of a system of spin-polarized fermionic molecules interacting via the dipole-dipole interaction while confined in a circular waveguide geometry and subjected to an additional static external field. We found that the Thomas-Fermi ground-state density distribution of the molecular fermions is periodic along the ring, with a large fraction of particles residing in the region of minimum energy. For small rings and large interaction strengths the particles are then localized in diametrically opposite regions. However, in the quasi-one dimensional case there is a lower bound to the number of particles for which that ground state is stable, and an upper bound in three-dimensional case. (The different scaling in these two situations is a direct consequence of the scaling of the kinetic energy of the fermionic sample with density.) The instability of the Thomas-Fermi ground state, and the concomitant breakdown of the Thomas-Fermi theory, signals the presence of nontrivial quantum correlations in the true many-body ground state, and a treatment beyond mean-field theory is needed to deal with that situation. It is known for instance that in quasi-one dimensional systems and for low particle densities fermionic systems can under appropriate circumstances undergo a Wigner crystallization, in analogy with the situation for trapped ions. We are now concentrating on the case of large, centimeter-size rings. Such rings, which are of considerable interest for practical applications, are also expected to result in situations far from the Thomas-Fermi situation. A quantum dynamical treatment using wave-packet methodologies is more appropriate in this case, and is currently under study. Once we have this development under control, hopefully in the next few months, we will be in a position to start evaluating quantitatively the pros and cons of using ultracold polar molecules in accelerometers and rotation sensors, a compared in particular with the use of neutral atoms.
Bibliography Type:	Description: (Last Updated: 02/24/2006)
Articles in Peer-reviewed Journals	Uys, H.; Miyakawa, T.;Meiser, D.; Meystre, P. "Passage time statistics in the formation of ultracold dimers from fermionic atoms" Phys. Rev. A 72, 053616 , Nov-2005
Articles in Peer-reviewed Journals	Dutta, O.; Jääskeläinen, M.; Meystre, P. "A polarizing beam splitter for dipolar molecules" Phys. Rev. A 71, 051601(R) , May-2005
Articles in Peer-reviewed Journals	Meiser, D.; Meystre, P. "Reconstruction of the phase of matter-wave fields using a momentum resolved cross-correlation technique" Phys. Rev. A 72, 023603 , Aug-2005

Articles in Peer-reviewed Journals	Miyakawa, T.; Meystre, P. "Anderson lattice description of photoassociation in an optical lattice" Phys. Rev. A 73, 021601(R) , Feb-2006
Articles in Peer-reviewed Journals	Jääskeläinen, M.; Meystre, P. "Coherence dynamics of Bose-Einstein condensates in asymmetric potentials" Phys. Rev. A 73, 013602 , Jan-2006
Books/Book Chapters	Search, C. P; Meystre, P. "Nonlinear and quantum optics of atomic and molecular fields" in "Progress in Optics Volume 47" Ed. Wolf, E. Elesevier, 139-214 , Sep-2005
Presentation	Meystre, P. "Quantum optics of molecules" invited talk, 14th International Laser Physics Workshop LPHYS'05, Kyoto, Japan Jul-2005
Presentation	Meystre, P. "Quantum optics of ultracold molecular fields" invited talk, Workshop on New Developments in Quantum Gases, Seattle WA Jul-2005
Presentation	Meystre, P. "Quantum optics of ultracold molecules" invited talk, Princeton-TAMU BEC Symposium, Princeton, NJ Oct-2005
Presentation	Meystre, P. "Quantum optics of ultracold molecular fields" Invited paper, International Conference on Quantum Optics, Hong Kong Dec-2005